

Tunisia, audited — the engineer’s ledger

the energy balance, water chemistry, acid-stream assays, lift floors, and every gate
— everything a technical team verifies before FID.

Provenance law: MEASURED = cited anchor · DERIVED = arithmetic on measured values · ASSUMED = declared default. commercially-validated = 0 — no buyer has signed anything; published as zero on purpose. Generated 2026-09-09 by nations_audit.py / make_nations_pdfs.py — the same code that renders the page.

1. The energy ledger, leg by leg

Recovery ledger recomputed: 33.9 GWh/yr recovered + 20 GWh/yr dispatch value. The printed 82 GWh did not sum from its own legs (7.2 + 30 + 20 = 57.2; 24.8 unexplained) — corrected, with the arithmetic, on the page.

LEG	FORMULA	VALUE
Pump-as-turbines	$9.81 \times (70\% \times 150,000/86,400) \text{ m}^3/\text{s} \times 50 \text{ m} \times 75\% \times 8,760 \text{ h}$	3.9 GWh/yr
Heat recovery	$0.6 \text{ GJ/t} \times 2 \text{ Mt PG} \rightarrow 333 \text{ GWh thermal} \times 9\%$	30 GWh/yr
Flexibility	desal + storage follow the solar curve	20 GWh/yr (dispatch, not joules)
ERD	inside the 4 kWh/m ³ SWRO benchmark (~35-40% of RO energy)	0 added — no double count

2. Water — the numbers an EPC checks first

PARAMETER	VALUE	ANCHOR
SWRO specific energy	4 kWh/m ³ (benchmark incl. ERD)	Rw3 (mega-plant class)
Lift physics floor	1.09 kWh/m ³ per 400 m at 100% eff; 1.45 at 75%	Rw4 — the physics that decides where water may travel
Delivered inland cost	\$1.10-1.50/m ³ vs irrigation tariffs \$0.04-0.07	Rw6/Rw7 — inland water runs on brackish, not long-haul
SWRO CAPEX	\$800-1,200 per m ³ /day (mega-plant)	Rw2
Gate	blended ≥\$0.68/m ³ , ≥60% pre-sold, brine permit	computed (DSCR)

3. Materials — the acid stream

PARAMETER	VALUE	ANCHOR
REE content, Gafsa rock	322 mg/kg avg, peaks >1,700	Rm1

PARAMETER	VALUE	ANCHOR
PG partition of REE	60-85%	Rm2
Acid-stream U	85-90% of rock-U partitions to the wet-process acid	Ru1 (Freeport precedent)
U recovery (proven)	>95% DEHPA/TOPO, 14M lbs over 1978-99	Ru1
Radiology	Ra-226 214-970 Bq/kg in PG; exemption threshold 1,000 Bq/kg	Rm11/Rm12
Phase 0 assays	12 stream points: U, REE, Ra-226, Po-210, full metals	itemized in the audit

4. The dependencies, by status

DEPENDENCY	STATUS
Desal brine permit	in design — Gabes outfall / diffuser; permit required before SWRO FID
PG chemistry (Ra-226 partitioning)	Phase 0 assay — flowsheet keeps radium in gypsum (Freeport precedent); re-verify on Gafsa stream
Uranium handling	gate — CNSTN licensing, NORM transport, export controls
REE separation capex	later phase — +\$244M solvent-extraction plant beyond the ~\$100M circuit
Solar bankability	gate — DSCR ~0.9-1.0 at the record-low tariff; bankable region or grant support required
Grid connection	later phase — 3 GW connection + 400 kV upgrades ~\$450M
Compute accelerators	gate — export-control licenses; colocation until an anchor tenant is licensed
Truffle yield	gate — 376 kg/ha is the 20-yr Murcia average; gate = 3-season record >=200 kg/ha
Export prices	exposure — U spot \$80-95/lb; truffle EUR20-40/kg — no contracted offtake yet
Financing	uncommitted — DFI channel modeled for solar/desal; nothing committed

The water stack — line-by-line, with each credit's failure mode

The \$0.347/m³ audit: the base is measured anchors; each credit is a loop-internal yield with a named failure mode; the degraded cascade still clears the \$0.68 gate.

LINE	\$/M ³	PROVENANCE	WHAT MAKES IT FAIL
Base — solar energy 4 kWh/m ³ at the loop's own PPA + capex + O&M	\$0.567	MEASURED anchors (Rc13/Rw2/Rw3)	holds — if even the PPA lapses, grid power takes the base to \$0.88, still under SONEDE
DC reject heat → RO feed preheat — one seawater pipe cools the servers, then arrives warm at the membranes (SEC -2.5%/°C)	-\$0.007	DERIVED — Leg 03 + Leg 01, same pipework	fails only if no compute anchor tenant — and costs \$0.007
Kiln/exotherm preheat — the acid plant and board kilns warm the same feed	-\$0.000	DERIVED — Leg 04/05 waste heat	negligible by design — carried at zero margin

LINE	\$/M ³	PROVENANCE	WHAT MAKES IT FAIL
Brine valorization — salt/Mg evaporated on the loop's own free heat, sold at the park gate (of-take signed; nothing hauled)	−\$0.137	DERIVED — brine is the desal's own by-product	if the of-take is not signed by FID, the credit degrades to −\$0.03 (disposal avoided only) — the biggest single sensitivity
Shared civil works — one intake, one outfall, one grid connection serves RO + DC + metals in the same park	−\$0.032	DERIVED — 12% capex sharing	if the park phases one module at a time, halve it to −\$0.016
Dispatch value — the desal's own flexibility follows the loop's own solar curve	−\$0.015	DERIVED — 20 GWh shifted	fails to zero if the STEG contract does not reward flexibility — contractual, not hoped
ESG financing delta — the health leg's public KPIs reprice the loop's own debt (100–150bp)	−\$0.026	DERIVED — nations_v3	fails to zero if lenders do not accept the KPI track record — bankable-region fallback keeps the module alive
Brine pump-as-turbine — the desal's own residual pressure	−\$0.002	DERIVED — 2.5 bar × 75%	fails to zero — immaterial
Integrated	\$0.347/m³	28% of SONEDE's \$1.00-1.50 (Pg16)	the degraded cascade: if every soft credit fails, water still lands at \$0.521/m³ — under SONEDE's own \$1.00–1.50, and the \$0.68 gate still clears with margin. The credits are the margin, not the plan.

The people's energy is electrons, not hydrogen — and the H₂ lives inside the loop

The loop's H₂ (2.63 kt/yr) is an industrial molecule: per kWh of heat, green H₂; at \$0.21 is 7× the loop's \$0.031/kWh electron; blending caps at 5-20% by volume (Pg20/Pg21/Pg22); the whole H₂; volume equals 0.0% of grid-gas demand. The molecule goes where only a molecule works: ammonia for the GCT chain, inside the loop. the electrolyzer sits on the desal line, and everything it touches stays in the loop: the water it splits is the SWRO permeate (0.04% of the 150,000 m³/d output); its un-split reject stream is clean water that returns to the product line; its oxygen by-product aerates the spirulina ponds and greenhouses; its stack heat (15–20% of input) joins the kiln and DC reject heat on the desal preheat circuit; and the minerals concentrated in its flush stream join the brine valorization line. The electrolyzer is not a consumer of the loop — it is another recovery point inside it.

The total routing — every stream gets a home

The ponds do *not* drink brine: SWRO brine is 55-70 g/L, past Arthrospira's ~30 g/L ceiling (Pg31); they drink the desal permeate and breathe the electrolyzer's oxygen. The full inventory:

STREAM	PRODUCED BY	ITS HOME	THE NUMBER
Solar electrons	500 MW fleet	desal, metals, DCs, electrolyzer, pumps — routed by the dispatcher	876 GWh/yr

STREAM	PRODUCED BY	ITS HOME	THE NUMBER
SWRO permeate	coastal desal	city offtake · electrolyzer feed · ponds · greenhouses	150,000 m ³ /d
SWRO brine	coastal desal	salt + magnesium valorization, evaporated on the loop's own free heat	122,727 m ³ /d → 58 kt Mg/yr
Brine bittern	valorization	Mg(OH) ₂ ; fire retardant → the board's own chemistry	140 kt/yr = 140% of the board's need
Electrolyzer oxygen	H ₂ ; plant	spirulina pond + greenhouse aeration	21 kt/yr
Electrolyzer hydrogen	H ₂ ; plant	ammonia → the GCT fertilizer chain	2.63 kt/yr
Electrolyzer reject water / heat / flush	H ₂ ; plant	product line / desal preheat / brine line	nothing leaves
DC reject heat	compute park	the same seawater pipe, arriving warm at the membranes	−\$0.007/m ³
Kiln + acid exotherm	board kilns + metals plant	desal preheat + greenhouses	30 GWh/yr
CO ₂ ; (kiln + power)	combustion streams	mineralized into board (2 Mt PG) · pond carbon	540 t CO ₂ /yr to the ponds
Phosphogypsum	GCT stack	fire-rated board — with its retardant from our own brine	2 Mt/yr
Acid stream	GCT wet-process acid	U + heavy REE extraction; raffinate back to fertilizer	716.5k lbs U ₃ O ₈ /yr
Municipal return flows	city mesh	30%+ reuse · pump-as-turbines on the descent	3.9 GWh/yr
Sludge / digestate	wastewater + organics	biogas for kilns and greenhouses	booked option, honestly labeled
Slaughterhouse bones	local slaughterhouses	bone char → the fluoride filters; spent char → phosphate soil amendment for the fields	filter loop closes into fertilizer
Spent pond medium	spirulina harvest	nutrient-rich → greenhouse fertigation	nothing discarded
Ammonia fertilizer	GCT chain	the terfas fields' host plants + greenhouse soils	the truffle grows on the loop's own nitrogen
Compute services	colo park	export — the data product	tenant-gated
Ra-226 / Po-210	acid stream radiology	stays in the gypsum/board matrix — the radiology gate	below the 1,000 Bq/kg exemption
Board offcuts	board line	recycled into new board	zero waste
Sodium chloride	brine line	industrial offtake at the park gate	nothing hauled
Halophytes / Artemia on brine	brine line	future option, honestly labeled — not a current credit	the only stream awaiting its consumer

the ponds drink the desal permeate — ~ 0.2 m³/d, 2% of the 150,000 m³/d output — and breathe the electrolyzer's oxygen; the brine (55–70 g/L, far past the 30 g/L ceiling the organism tolerates) is routed to salt and magnesium valorization instead. Halophytes and Artemia on brine are booked as a future option, honestly labeled — not a current credit.

The fuel invoice, honestly split — the 500 MW is the grid's medicine, not the whole prescription

of the \$4.895B fuel invoice, the 500 MW fleet displaces only the power-sector gas slice — $\sim \$0.75$ B/yr (DERIVED from the energy balance) — by \$106-125M/yr, which is 14133-16667% of that slice. The rest of the invoice has its own answers, not solar's: the bottles ($\sim \$0.55$ B of LPG, the bonbonne) leave the market as the people shift to electrons — induction on the loop's own surplus — and the biogas pilot; transport fuels ($\sim \$2.6$ B) are the honest frontier. The 500 MW is the grid's medicine, not the country's entire prescription.

The mountains, honestly analyzed

Fog collection is real where fog lives (10-20 L/m²/day, Pg33) — Tunisia's fog lives in the already-humid NW; the truffle mountains are the arid interior, and cloud seeding has no proven routine gain (Pg32). What the mountains give the loop: the mountains enter the loop as storage, rainfall and free cooling — and as the land where the truffle belongs (freeing the lowlands that grow today's food). Fog harvesting from pipe meshes stays off the program: the moisture is not there to collect, and the honest scientist does not plant a net under a cloudless sky.

What this document is: a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

What it is not: a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

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